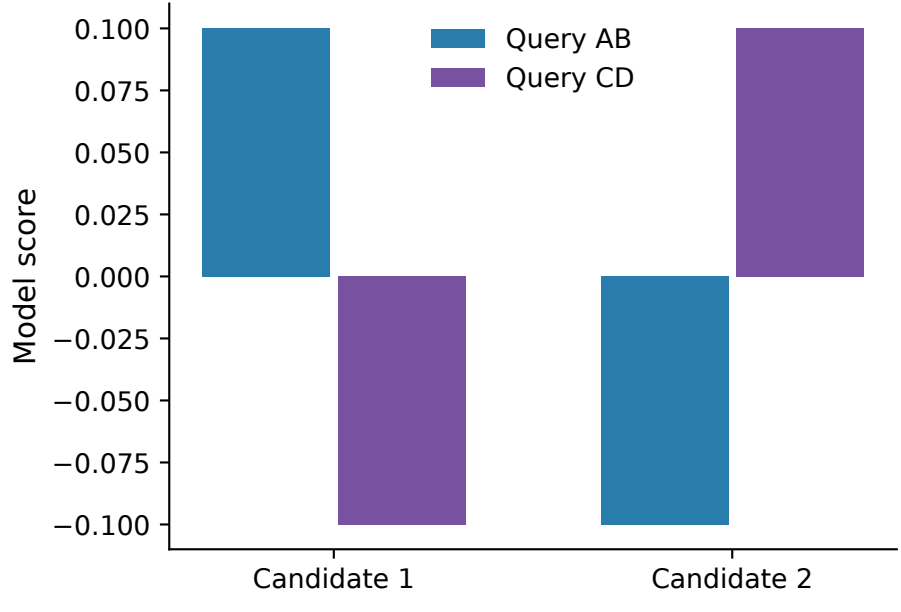
(a) Identical coordinatewise statistics

Statistic	Query AB	Query CD
Mean	[0.0, 0.0]	[0.0, 0.0]
Second moment	[1.0, 1.0]	[1.0, 1.0]

(b) Different pair-identity predictions